Data Communication & Computer Networks (CN)

Topic 7 of 7 : Network Security

Curation in progress — 35 auto-extracted questions pending verification (see Pending Review at the end)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** verified MCQs + **35** pending review

Question Type: MCQ — every question has 4 options (A, B, C, D)

Languages: **Bilingual:** English + Hindi (हिन्दी)

Every question includes: ✔ Correct answer (why it is RIGHT) • ★ Nukta / key point (exam tip) • ✖ why each wrong option is WRONG

Verification: Every answer cross-verified **4 times** (Forouzan facts + official answer keys of Subject Test / Unit Test Solutions where available)

Syllabus Coverage — sub-topic wise order

Sub-topic curation in progress — auto-extracted questions are listed in the **"Pending Review"** section at the end.

How to read this PDF: Each question shows 4 options (A-D) in English and Hindi. The line starting with ✔ gives the correct answer and why it is right. The line starting with ★ is the **Nukta** — the key point to remember for the exam. The lines starting with ✖ explain why every wrong option is wrong. Read the question, try it yourself, then check the answer.

System: This PDF is generated by **tools/cn_build_topic.py** from **tools/data/topic7_data.py**. To add more questions, drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions are auto-classified into topics and appended here (as pending) for review. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

These questions were auto-extracted & classified from uploaded PDFs. Answers/nukte to be verified before counting.

⚠ NEW — auto-ingested, answer needs verification (pending review)

P1. Network Attacks & Description 9 List–I (Attack

(A) Phishing ( )

(B) DoS Attack

(C) Man-in-the-Middle

(D) Spoofing (R) List–II (Description /) (i) Flooding network traffic / 9 (ii) Stealing user credentials (iii) Intercepting communication B W+ (iv) Impersonation attack Codes / : (A) a–ii, b–i, c–iii, d–iv (B) a–i, b–ii, c–iii, d–iv (C) a–iii, b–i, c–ii, d–iv (D) a–ii, b–iii, c–i, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P2. In the digital signature technique when the whole message is signed using an asymmetric key, the receiver of the message uses _____ to verify the signature. v B Asymmetric Key W + , - y W + z _____

(A) Her or his own public key

(B) Her or his own private key

(C) The sender’s public key

(D) None of these

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P3. The text that gets transformed using an algorithm cipher is called: " $ 2 $: ;

(A) Complex Text

(B) Transformed Text

(C) Plain Text

(D) Scalar Text

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P4. In cryptography, which statement about Plaintext, Ciphertext and Key is MOST accurate? ‰Š ‹ B \ Œ, Œ

(A) Plaintext is generated after decryption using a public key

(B) Ciphertext is the original readable message

(C) Plaintext is transformed into Ciphertext using an algorithm and a key

(D) Ciphertext and plaintext are always of the same length

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P5. Which attack becomes possible if the same key is reused multiple times in symmetric encryption

(A) Replay Attack
(B) Known Plaintext Attack
(C) Man-in-the-Middle Attack
(D) Phishing Attack

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P6. Data Encryption Standard (DES) ; ?Data Encryption Standard (DES) is a:

(A) Block Cipher
(B) Stream Cipher
(C) Bit Cipher
(D) Byte Cipher

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P7. In the virus life cycle, which phase actually performs the malicious activity? H B - h

(A) Infection Phase / Z
(B) Triggering Phase
(C) Dormant Phase / C
(D) Execution (Payload) Phase / F

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P8. Which of the following backup media provides Write Once Read Many (WORM) functionality? k l B Write Once Read Many (WORM) @

(A) Hard Disk / h 1
(B) Magnetic Tape / j
(C) Optical Disk (CD-R / DVD-R)
(D) Flash Drive / k 9

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P9. IPSec operates in which mode(s)? IPSec B P

(A) Transport mode only / 9 h
(B) Tunnel mode only
(C) Transport mode and Tunnel mode / 9 h
(D) Application mode / n$

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P10. SSL/TLS handshake protocol is mainly used for:SSL/TLS u @ v w x ;

(A) Data encryption
(B) Data compression
(C) Key exchange and authentication
(D) Packet routing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P11. IPSec is MOST commonly used to implement: IPSec

(A) Secure Email
(B) Secure Web Browsing
(C) Virtual Private Network (VPN)
(D) Wireless Security

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P12. In IPSec Tunnel mode:/ IPSec B ;

(A) Only payload is encrypted
(B) Only header is encrypted
(C) Entire IP packet is encrypted
(D) No encryption is used

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P13. Which of the following uses the same key for encryption and decryption? H I + - I}y~ y~

(A) Symmetric Key Cipher
(B) Asymmetric Key Cipher
(C) Hash Function
(D) Digital Signature

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P14. Which of the following uses a pair of keys (public and private)? H I + (9)

(A) Symmetric Cipher
(B) Asymmetric Cipher
(C) DES
(D) AES

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P15. Which of the following is faster in encryption and decryption? I}y~ y~ + -

(A) Symmetric Key Cipher
(B) Asymmetric Key Cipher
(C) Can not decide
(D) None of the these

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P16. Public key is used to _____. I• _____

(A) Encrypt the data
(B) Decrypt the data
(C) Both encrypt and decrypt
(D) Generate session keys

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P17. Which of the following is NOT a symmetric encryption algorithm? H I I}y~ € d •

(A) AES
(B) DES
(C) Blowfish
(D) RSA

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P18. What is the main disadvantage of symmetric key encryption? h I}y~ T

(A) It is very slow
(B) Key distribution is difficult
(C) It uses two keys
(D) It cannot encrypt text

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P19. In public key cryptography, the message encrypted with the public key can only be decrypted by _____. I• y, m +, I• I}y, _____ ƒ

(A) Another public key
(B) Same public key
(C) Private key corresponding to that public key
(D) Session key

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P20. Which one of the following statements is true about symmetric and asymmetric cryptography? h h y, m + H + -

(A) Symmetric cryptography provides authentication.
(B) Asymmetric cryptography is faster than symmetri
(C) C. Symmetric cryptography requires secure key exchange.
(D) Both use two different keys.

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P21. What is the key length of AES-256 encryption? AES-256 I}y~

(A) 128 bits
(B) 192 bits
(C) 256 bits
(D) 512 bits

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P22. Which cipher method encrypts one bit or byte at a time? I}y,

(A) Block Cipher
(B) Stream Cipher
(C) Substitution Cipher
(D) Transposition Cipher

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P23. Which of the following statements is FALSE? H + -

(A) AES is a block cipher.
(B) RSA is a symmetric cipher.
(C) Diffie-Hellman is used for key exchange.
(D) DES is symmetric. Which of the following is NOT a goal of network security? H I + - 9 t „ ? A. Confidentiality ( ) B. Integrity ( ) C. Availability (Y) D. Redundancy ( )

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P24. The term "CIA triad" in information security refers to _____. t + "CIA] " _____

(A) Cryptography, Integrity, Authentication
(B) Confidentiality, Integrity, Availability
(C) Confidentiality, Information, Access
(D) Control, Information, Access

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P25. The goal of “Integrity” ensures that

(A) Data can be accessed only by authorized users

(B) Data is accurate and has not been modified

(C) Data is always available when needed

(D) Unauthorized users cannot enter the system 40 Which of the following attack violates confidentiality? H I + j ? A. Interception B. Modification C. Fabrication D. Interruption

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P26. Modification attack violates which security goal? t „ j

(A) Confidentiality

(B) Integrity

(C) Availability

(D) Authentication

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P27. Denial of Service (DoS) attack violates which goal? Denial of Service (DoS) „ j

(A) Confidentiality

(B) Integrity

(C) Availability

(D) Authentication

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P28. Spoofing attack targets which type of security aspect? ˆ t t

(A) Authentication

(B) Integrity

(C) Availability

(D) Confidentiality

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P29. Which of the following is an active attack? H I + - y (Active)

(A) Interception

(B) Modification

(C) Traffic Analysis

(D) Eavesdropping

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P30. The Caesar Cipher is an example of which type of cryptographic technique? Caesar Cipher y, m

(A) Transposition Cipher
(B) Substitution Cipher
(C) Stream Cipher
(D) Block Cipher In Steganography, the file used to hide secret data is called _____. Steganography + + p A. Secret File B. Cover File C. Cipher File D. Key File

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P31. The process of detecting hidden data inside a file is known as _____. ? y

(A) Cryptanalysis
(B) Steganalysis
(C) Data Mining
(D) Stegnography

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P32. What is the output of a Steganography process called? Steganography ? y

(A) Ciphertext
(B) Stego-object
(C) Encoded file
(D) Key file

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P33. Which of the following is used in HTTPS for encryption?HTTPS

(A) SSL/TLS
(B) FTP
(C) ARP
(D) NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P34. What type of NAT maps multiple private IPs to one public IP? NAT IP IP

(A) Static NAT
(B) Dynamic NAT
(C) PAT (Port Address Translation)
(D) Double NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P35. Which mode of IPsec provides end-to- end security?IPsec

(A) Transport mode

(B) Tunnel mode

(C) Static mode

(D) Gateway mode What type of attack is a Smurf attack?Smurf (a) DoS (b) Spoofing (c) Phishing (d) Brute force

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]